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Carrier mobility in organic semiconductors with an exponential density of states under the framework of admittance spectroscopy

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ABSTRACT

The carrier transport in amorphous organic semiconductors with an exponential density of states is systematically studied under the framework of admittance spectroscopy. Due to the exponential distribution, the slope of the double logarithmic curve of current density versus voltage is expressed as b+2, where b relates to the width of the density of states. And the mobility is proportional to n^b, where n is the density of carrier. In this case, the peak frequencies of the negative differential susceptance and imaginary part of impedance turn out to be functions of parameter b. Therefore, the relation between transit time and mobility will be a linear function of parameter b. Applying our model to interpret experimental data of both small molecular and polymeric materials are illustrated. The contributions of electric field and of carrier density to mobility are discussed.

1. Introduction

Organic materials have been widely studied for their potential use in organic semiconductor devices such as organic light-emitting diodes (OLEDs), field-effect transistors, and solar cells. Understanding the charge transport properties is important to improve the device performances. One of the most important parameters that determines the device performances is the carrier mobility μ.

Admittance spectroscopy (AS) based on small signal space-charge-limited current (SCLC) theory is one of the most widely used techniques to characterize the carrier dynamics in organic materials [1–5]. The theoretical study of AS for space-charge-limited (SCL) condition can be date back to 1960s. In 1961, J. Shao and G. T. Wright first derived an AS expression for the case of SCL diodes with a constant mobility [6]. In 1966, D. Dascalu generalized Shao's theory to the case of an electric field dependence of mobility [7]. But these early works did not discuss techniques measuring mobility with AS.

In 1999, H. C. F. Martens et al. studied the frequency-dependent electrical response of holes in poly(p-phenylene vinylene) (PPV) [1]. The authors gave a paradigm to measure carrier mobility using AS for the constant mobility under SCL condition. Later, the authors advanced the negative differential susceptance (-ΔB) method for organic materials mobility measurement, in which a frequency of the maximum of -ΔB is related to the mobility [2]. In 2006, S. W. Tsang et al. derived a similar AS expression and measured the mobilities of several small molecular organic materials [8]. One drawback of the -ΔB method is that the measurement would be of difficulties in determination of mobility in dispersive transport [9,10]. To overcome this difficulty D. C. Tripathi et al. suggested that the frequency dependence of imaginary part of impedance (ImZ method) could be a simpler, more convenient, and powerful method of determination of mobility [10]. Up to date, these two methods have been widely used to characterize transport properties in organic materials [11,12].

It is worth noting that, for most of the organic materials, the mobility is not a constant but a function of carrier density [13–15] and/or electric field [16–19]. The dependence of the mobility on the carrier density or on the electric field can result in an enhancement of the slope of the current density versus voltage (J-V) curve. However, in some experiments, starting from the well-known Mott-Gerney (MG) AS model, a Poole Frenkel field effect, i.e. $\ln(\mu)\sim\sqrt{E}$, was concluded [8,20]. What's more, in some publications, carrier mobility was measured without a discussion of the slope of J-V curve [21]. C. Tanase et al. suggested that the enhancement of the SCLC for some polymers at high bias at room temperature is due to the carrier density dependence of the hole mobility [22]. While the dependence of the mobility on the electric field was only observed at low temperatures [14,22,23]. Therefore, an AS mobility model including the carrier density dependence of the mobility is imperative.

In 1998, M. C. J. M. Vissenberg and M. Matters advanced a percolation mobility expression for organic materials with exponential

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density of states (EDOS) [13]. They showed that in this condition the mobility should be a power function of the carrier density, i.e., $\mu = an^b$, where a is a material and temperature dependent pre-factor, and $b = T_0/T - 1$ ($b > 0$), with T_0 the width of the density of states (DOS). For this special case, B. Ramachandhran et al. presented an exact analytical ac admittance expression, suggests that the b dependent critical response frequency may be different from the inverse of the carrier transit time [24]. That means, for this kind of materials, the mobility may not be simply proportional to the peak frequencies of -ΔB and ImZ.

In this work, following the study of B. Ramachandhran et al., we discuss the feasibility of measuring mobility using AS for the case of EDOS. The proportional coefficients between peak frequencies and the mobility are calculated for -ΔB and ImZ methods. Analysis suggests that the transit time and the frequency would be a function of parameter b, which can be obtained from J-V curves. Most importantly, the mobility value may increase as the bias rises because an increment of the carrier density. This phenomenon may cause illusion as if the mobility may be a function of electric field, which is in contradict to the MG theory assumption that mobility is not a function of electric field. Hence, a discussion of Poole-Frenkel electric field effect of mobility may not be objective.

2. Theory

2.1. Frequency dependence of the admittance and mobility of MG SCLC model

To measure the mobility, organic materials should be prepared in single carrier device structure with at least one Ohmic contact to form SCLC under bias. If the mobility is constant and the device is trap-free, the current density can be written as the well-known MG law:

$$J_{\mathrm{MG}} = \frac{9}{8}\mu\varepsilon\frac{V^2}{L^3}, \tag{1}$$

with μ the carrier mobility, ε the dielectric constant of the organic material, V the operating voltage, L the device thickness. From Eq. (1) the value of μ can be derived from the measured J-V curve directly. This is an ideal situation, and only a small number of materials exhibit constant mobilities. For most organic materials, the mobility of carrier will be a function of carrier density as discussed in the next section.

Alternatively, we can determine the mobility by measuring AS of single carrier device if the mobility is a constant. The AS expression can be obtained under SCLC condition. If a small amplitude ac voltage signal, $v_{\mathrm{ac}} = V_{\mathrm{m}}\sin(\omega t)$, is applied to the device discussed here, a current signal j_{ac} will be generated. The ratio of the j_{ac} to the v_{ac} is the admittance $Y(\omega) = G + \mathrm{i}\omega C$, with $\mathrm{i}^2 = -1$, $\omega = 2\pi f$ the angular frequency, f the frequency, G the conductance, and C the capacitance. J. Shao and G. T. Wright derived the incremental admittance of the SCL dielectric diode as:

$$Y(\Omega) = G + i\omega C = \frac{g\Omega^3}{6}\left[\frac{\Omega - \sin\Omega}{(\Omega - \sin\Omega)^2 + \left(\frac{\Omega^2}{2} + \cos\Omega - 1\right)^2} + \mathrm{i}\frac{\cos\Omega - 1 + \frac{\Omega^2}{2}}{(\Omega - \sin\Omega)^2 + \left(\frac{\Omega^2}{2} + \cos\Omega - 1\right)^2}\right], \tag{2}$$

with $\Omega = \omega \times \tau$, τ the transit time, g the steady-current incremental conductance.

Fig. 1a shows that the calculated capacitance is a function of frequency according to Eq. (2). At high frequencies, the applied ac signal cannot redistribute the space charge and hence the capacitance will be the geometric capacitance C_{geo}. Apparently, at low frequencies, a negative contribution to the device capacitance appears. As predicted by Eq. (2), the saturation capacitance value at low frequency would be $0.75C_{\mathrm{geo}}$. This negative contribution is an inductive capacitance stems from the finite transit time τ of injected carriers. The characteristic frequency where this contribution disappears is not easy to be distinguished.

However, due to this negative contribution, a maximum will appear in the negative differential susceptance curve, as shown in Fig. 1c. H. C. F. Martens et al. presented the negative differential susceptance -ΔB method (-$\Delta B = \omega[C - C_{\mathrm{geo}}]$) to determine the carrier mobility. The plot of -ΔB peaks at a frequency $f_{\max}$, which scales with the corresponding transit time. The authors gave an approximate relation between τ and $f_{\max}$, $\tau \approx 0.72 \times f_{\max}^{-1}$ [25]. The equation for τ and μ in SCL condition is given by:

$$\mu = \frac{4}{3}\frac{L^2}{(V - V_{\mathrm{bi}})\tau}, \tag{3}$$

with V_{bi} the built-in voltage. Combining Eq. (3) and the approximate relation, the expression for μ can be easily obtained:

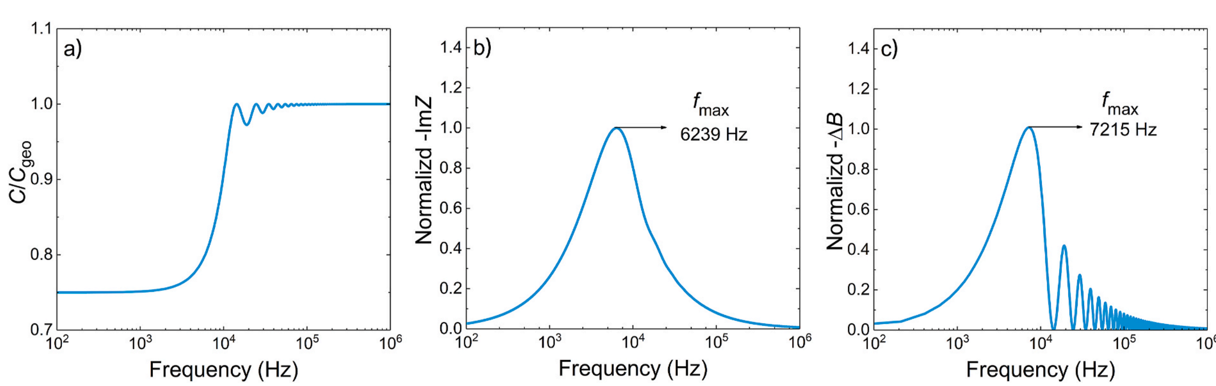

Fig. 1. (a)The simulated normalized capacitance (C=ImY/ω) in the MG-SCLC model, calculated from Eq. (2) for a transit time $\tau = 10^{-4}$ s. (b) The normalized -ImZ and peak frequency $f_{\max}$. (c) The normalized -ΔB and peak frequency $f_{\max}$ calculated from capacitance.

$$\mu = \frac{4L^2}{3(V - V_{\mathrm{bi}})}\frac{f_{\max}}{\eta}, \tag{4}$$

with η the proportionality coefficient. By plotting the change in susceptance and determining the corresponding $f_{\max}$, the value of μ can be calculated from Eq. (4).

If the charge transport exhibits dispersive characteristics, calculation of difference susceptance may not be possible under certain circumstances since it involves a subtraction relative to a flat baseline of C_{geo} which itself in certain cases becomes difficult to obtain. In fact, no peak can be found in case of even moderate dispersion where the characteristic dip in the Capacitance curve does not occur.

To overcome this difficulty, Y. N. Mohapatra et al. illustrated that the frequency dependence of imaginary part of impedance (ImZ) could be a powerful and convenient tool to determine carrier mobility. By using a proportional coefficient of about 0.47 between the ImZ peak frequency $f_{\max}$ and τ, they confirmed the mobilities of m-MTDATA and polymeric MEH-PPV with other techniques [10]. The value of μ can be calculated by obtaining the peak frequency of the -ImZ [11], similar with -ΔB method. For highly dispersive transport, the ImZ peak can still be observed and quite stable. N. D. Nguyen et al. also specified the experimental limitations of -ΔB method in determining mobility [9]. Nonetheless, both ImZ and -ΔB methods can be served to obtain the materials' mobility.

The peaks of ImZ and -ΔB and the $f_{\max}$ can be observed in Fig. 1b and c. The corresponding proportionality coefficient η for each of ImZ and -ΔB can also be obtained are 0.62 and 0.72, respectively. The simulation result of η for -ΔB obtained here in Fig. 1c are similar to that of Ref. [25]. It is noteworthy that the mobility and transit time relation sometimes is given as $\mu = L^2/(V\tau_{\mathrm{dc}})$, see Refs. [1,8,10]. Thus, the τ_{dc} to $f_{\max}^{-1}$ ratio obtained using ImZ method by Y. N. Mohapatra et al. [10] is 0.75 of the simulation result η herein.

2.2. Frequency dependence of the admittance of EDOS model

For ideal materials, the mobility can be obtained directly from the slope of the SCLC of MG model. But for most of the amorphous organic materials, Monte Carlo simulations by Bässler et al. showed that the charge transport was characterized by a hopping process between localized sites in disordered materials [26]. These sites are located in different environments with an energy distribution, i.e., DOS [27–30]. The mobility is strongly dependent on the shape of DOS. Although the "real" DOS changes from material to material and usually has a complex shape, conventionally it is approximated by a Gaussian function whose parameters characterize the organic semiconductor [27,30–33]. Straightforward derivation of mobility expression from Gaussian distribution is difficult because the integration of Gaussian function cannot be work out analytically [14]. However, as reported in Ref. [13], an exponential DOS is a good approximation of the tail states of the Gaussian distribution that represent the most effective part of the distribution itself [30,32].

For the specific case of an exponential density of states (EDOS), Vissenberg and Matters showed that the mobility depends on the carrier density:

$$\mu_{\mathrm{VM}} = an^b, \tag{5}$$

where a is a material- and temperature-dependent pre-factor, and where $b = T_0/T-1$, $b > 0$ with T_0 the width of the EDOS [13].

On the basis of the EDOS model proposed by Vissenberg and Matters, B. Ramachandhran et al. derived a J-V relationship, incorporating the influence of charge density on mobility, as follows [24]:

$$J = a\varepsilon\left(\frac{\varepsilon}{\mathrm{q}}\right)^b \frac{(b+1)^{b+1}(2b+3)^{b+2}}{(b+2)^{2b+3}}\frac{V^{b+2}}{L^{2b+3}}. \tag{6}$$

This expression can be regarded as a generalization of Eq. (1) for arbitrary b and can be simplified to the MG law under the premise that $b = 0$. Using this current density expression, the electric field and density distribution will depend on parameter b:

$$E(x) = \frac{2b+3}{b+2}\frac{V}{L}\left(\frac{x}{L}\right)^{\frac{b+1}{b+2}}, \tag{7}$$

and

$$n(x) = \frac{\varepsilon}{\mathrm{q}}\frac{(b+1)(2b+3)}{(b+2)^2}\frac{V}{L^2}\left(\frac{x}{L}\right)^{\frac{-1}{b+2}}. \tag{8}$$

The transit time τ derived from Eq. (7) and Eq. (8) will also be a function of parameter b:

$$\tau = \int_0^L \frac{\mathrm{d}x}{\mu(x)E(x)} = \frac{(b+2)^{2b+2}}{(b+1)^{b+1}(2b+3)^{b+1}}\frac{1}{a}\left(\frac{\mathrm{q}}{\varepsilon}\right)^b\frac{L^{2b+2}}{V^{b+1}}. \tag{9}$$

Since the μ_{VM} takes into account the effect of density n, carrier mobility cannot be derived directly from the double logarithmic J-V curve as in the case of the MG law. The corresponding AS model is also required for determining μ_{VM}. The corresponding AS expression, required to determine μ_{VM}, presented by Ramachandhran et al. is as follows:

$$Y(\Omega) = \mathrm{Re}\left((2b+3)\frac{1}{v^*(\Omega)}\right)A\frac{dj_{\mathrm{ac}}}{dV} + \mathrm{Im}\left(\frac{1}{\Omega v^*(\Omega)}\right)C_{\mathrm{geo}}, \tag{10}$$

with A the device area. The dimensionless voltage function $v^*(\Omega)$ is obtained by integrating the dimensionless field function $e^*(\xi,\Omega)$ over the device thickness as follows:

$$\begin{aligned} v^*(\Omega) &= \int_0^1 e^*(\xi,\Omega)d\xi \\ &= \int_0^1 \frac{1}{b+2}\xi^{\frac{-1}{b+2}} exp\left(\frac{-\mathrm{i}\Omega}{b+1}\xi^{\frac{b+1}{b+2}}\right)\left(\int_0^\xi exp\left(\frac{\mathrm{i}\Omega}{b+1}x^{\frac{b+1}{b+2}}\right)dx\right)d\xi. \end{aligned} \tag{11}$$

As the analytical formula for $v^*(\Omega)$ cannot be derived, the solution of the $Y(\Omega)$ is calculated numerically here.

Fig. 2 illustrates that the $C(f)$, -$\Delta B(f)$ and Im$Z(f)$ plots with selected b values. To the best of our knowledge, T_0 are typically between 300 K and 550 K [32,34]. Therefore, we consider meaningful value of parameter b in the small range of 0–2. In low frequency range, as shown in Fig. 2a, the capacitance is given by:

$$C(0) = \frac{2b+3}{3b+4}C_{\mathrm{geo}}. \tag{12}$$

For $b = 0$, the capacitance $C(0) = \frac{3}{4}C_{\mathrm{geo}}$, consistent with the results of the MG SCLC model. For $b>0$, $C(0)$ is smaller than C_{geo}. As the value of b increases, the process of C moving from a low-frequency regime to oscillating high-frequency regime is prolonged. Fig. 2b and c exhibit the effect of b on the peaks ($f_{\max}$) of -Im$Z(f)$ and -$\Delta B(f)$ calculated with the same transit time τ. An increase in the b-value causes a reduction of the -ImZ amplitude and an increment of the -ΔB amplitude. However, the peak frequencies of both methods increase similarly due to the growth of b-value. The dependence of peak frequencies on parameter b indicating that the distribution of DOS may influence the transient time as expected by the density dependence of mobility.

With the respective peak frequencies of -$\Delta B(f)$ and -Im$Z(f)$ corresponding to b in Fig. 2, it is possible to fit an approximate linear relation of the coefficient η_b and b in the EDOS model, as presented in Fig. 3.

For the -ΔB method, the fitted linear equation for τ and $f_{\max}$ is:

$$\tau \approx (0.72 + 0.65b) \times f_{\max}^{-1}, \tag{13}$$

and for the ImZ method is:

$$\tau \approx (0.62 + 0.49b) \times f_{\max}^{-1}. \tag{14}$$

The η_b results for -ΔB and ImZ method are both consistent with MG

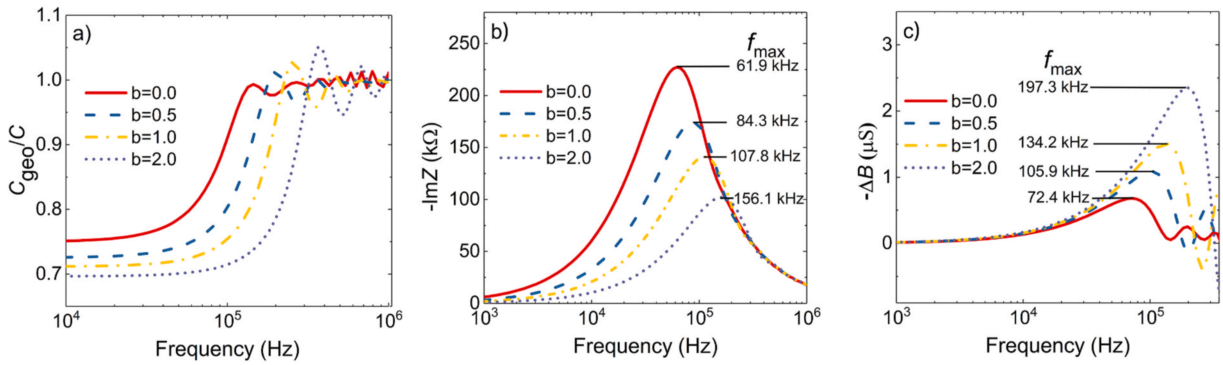

Fig. 2. The simulated frequency dependence of normalized C, -ΔB and ImZ in the EDOS model. The simulated transit time $\tau = 10^{-5}$ s and the range of b is chosen from 0 to 2. It can be seen that for the same transit time, the value of b also has an effect on the peaks or $f_{\max}$ of ImZ and -ΔB.

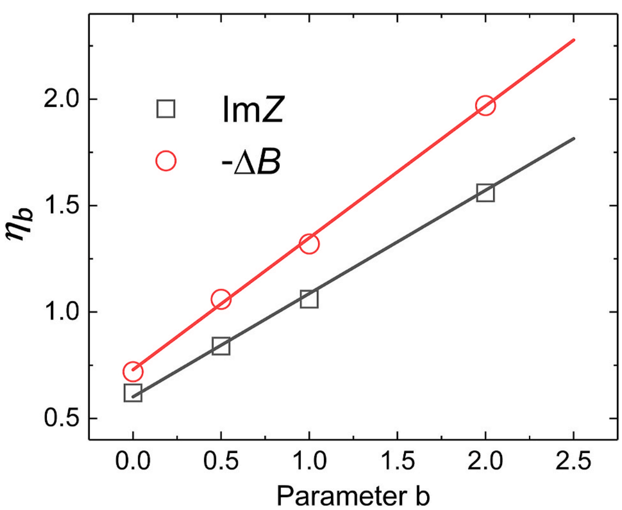

Fig. 3. The plot of proportion coefficients η_b (symbols) versus b for the ImZ method and the -ΔB method. The two solid lines are their linearly fitted straight lines respectively.

model coefficients η when the value of $b = 0$. Apparently for arbitrary $b > 0$, the scale factor η_b of the EDOS model is consistently larger than that of the MG model η. The dependence of $f_{\max}$ on parameter b, results in a small mobility value of EDOS situation compared with that of MG situation, as discussed below.

3. Discussion

3.1. Carrier mobility of EDOS model

Ramachandhran et al. did not continue to specifically study the carrier mobility of the EDOS model in Ref. [24]. In this paper, on the basis of their work, the formula of mobility in EDOS model is derived by combining Eq. (5), Eq. (8) and Eq. (9):

$$\mu_{\mathrm{VM}}(x) = an^{b} = \frac{(b+2)^2}{(b+1)(2b+3)} \frac{L^2}{V}\left(\frac{x}{L}\right)^{\frac{-b}{b+2}} \tau^{-1}. \tag{15}$$

Since μ_{VM} in Eq. (15) is a function of position x with $b > 0$, a numerical manipulation can be performed to obtain the average mobility

$$\mu_{\mathrm{VM}} = \frac{\int_0^L \frac{(b+2)^2}{(b+1)(2b+3)} \frac{L^2}{V}\left(\frac{x}{L}\right)^{\frac{-b}{b+2}} * \tau^{-1} \mathrm{d}x}{L} = \frac{(b+2)^3}{2(b+1)(2b+3)} \frac{L^2}{V}\tau^{-1}. \tag{16}$$

Combined with the relation between the η_b and b obtained from Fig. 3, i.e., Eq. (13) and Eq. (14), μ_{VM} can also be written as an equation for the peak frequency:

$$\mu_{\mathrm{VM}} = \frac{(b+2)^2}{2(b+1)(2b+3)} \frac{L^2}{V} \frac{f_{\max}}{\eta_b}. \tag{17}$$

When interpreting experimental data, the value of b is obtained from the slope of double logarithmic J-V curve. And the average mobility value can be calculated according to Eq. (17).

To study the influence of b on mobility, we calculate the average mobility as a function of b, as displayed in Fig. 4a. Obviously, the μ_{VM} value is always lower than that of the MG model for the same $f_{\max}$. Fig. 4b illustrates the relative error δ between the simulated mobility

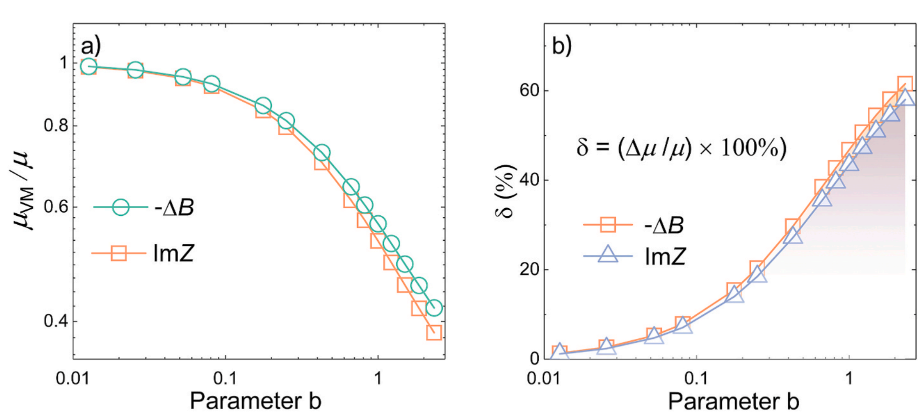

Fig. 4. (a) The plot of simulated μ_{VM}/μ for the same peak frequency $f_{\max} = 1620$ Hz. (b) Relative error plot of the mobility of the two models calculated from panel (a).

calculated by the two models. There is no significant difference between the simulated μ_{VM} results obtained by using the ImZ method or the -ΔB method.

According to Eq. (6), a slight change of b will not influence the slope of double logarithmic *J-V* curve. While the relative error δ of mobility can be up to 20 % with b of 0.2, as shown in Fig. 4a. Unfortunately, in literature some discussions did not discriminate between MG-SCLC and EDOS situations. And in some publications even *J-V* data were not given. When the value of b is greater than unit, there may be a strongly overestimation of mobility up to 3 times. This overestimation may essentially arise from the effect of carrier density on the frequency response of the AS. Compared to the AS of the MG-SCLC device, the peak frequency f_{max} obtained by the AS of the EDOS device may be higher for the same transit time.

3.2. Carrier density dependence

For most of the amorphous organic materials, carrier mobility may be a function of carrier density and electric field. Usually, researchers will systematically study the mobility under different electric field, i.e., Poole-Frenkel effect [16–19]. However, C. Tanase et al. suggested that the enhancement of the SCL hole current of PPV derivatives at high bias at room temperature is due to the carrier density dependence of the hole mobility. While the dependence of the mobility on the electric field is only observed at low temperatures [14,22,23]. Therefore, we discuss the density dependence of the mobility, as depicted by the Vissenberg and Matters model. To make the calculation more meaningful, we use the specific form of the mobility formula:

$$\mu_{VM} = an = \frac{\sigma_0}{q}\left[\frac{\sin\left(\pi\frac{T}{T_0}\right)}{B_c}\frac{T_0^4}{(2\alpha)^3T^4}\right]^{\frac{T_0}{T}} \times n^{\left(\frac{T_0}{T}-1\right)}, \tag{18}$$

with σ_0 the pre-factor for the conductivity, α^{-1} the effective overlap parameter between localized states, B_c the critical number for the onset of percolation. For a three-dimensional amorphous system [35], $B_c = 2.8$. With this mobility expression, the slope of current density would be influenced by parameter b as expressed in Eq. (6). Fig. 5a exhibits three current density curves with selected values of parameter b. The other parameters chosen here are $T_0 = 400$ K, $\sigma_0 = 1.3 \times 10^5$ S/m and $\alpha^{-1} = 0.26$ nm, which was successfully tested on hole-only diode devices of poly[2,5-bis(3′,7′-dimethyloctyloxy)-p-phenylene vinylene] ($OC_{10}C_{10}$-PPV) [36]. Dash line in Fig. 5a is a MG plot for slope comparison. It should be noted that for the MG case, the mobility-electric field plot the mobility is essentially a constant. However, for the case of EDOS, mobility will appear as if a function of electric field. In Fig. 5b. While it is worthy to note, the constant slope of *J-V* plot is not a fingerprint of the electric field dependent mobility.

This would lead one to the artificial Poole Frenkel effect, as did in some publications. But the assumption of our EDOS model is that the mobility is only a function of carrier density. As emphasized by C. Tanase et al., an increase of the applied bias gives rise to a simultaneous increase of the electric field and charge carrier density [22,32]. Consequently, the contributions of the charge carrier density and the electric field to the mobility cannot be disentangled from a SCL current. However, using the transistor device struture, these contributions can be separately studied. In their work they demonstrated that the enhancement of the current density in SCL diodes is dominated by the carrier density dependence of the hole mobility at room temperature and, in contrast to what has been assumed so far, not by its field dependence. Without a consideration of density dependence will lead to a strong overestimation of the field dependence of the mobility.

In order to quantify the amount of charge that flows through a EDOS device, we will use the average charge density n_{av} that is calculated from the distribution of $n(x)$, given by:

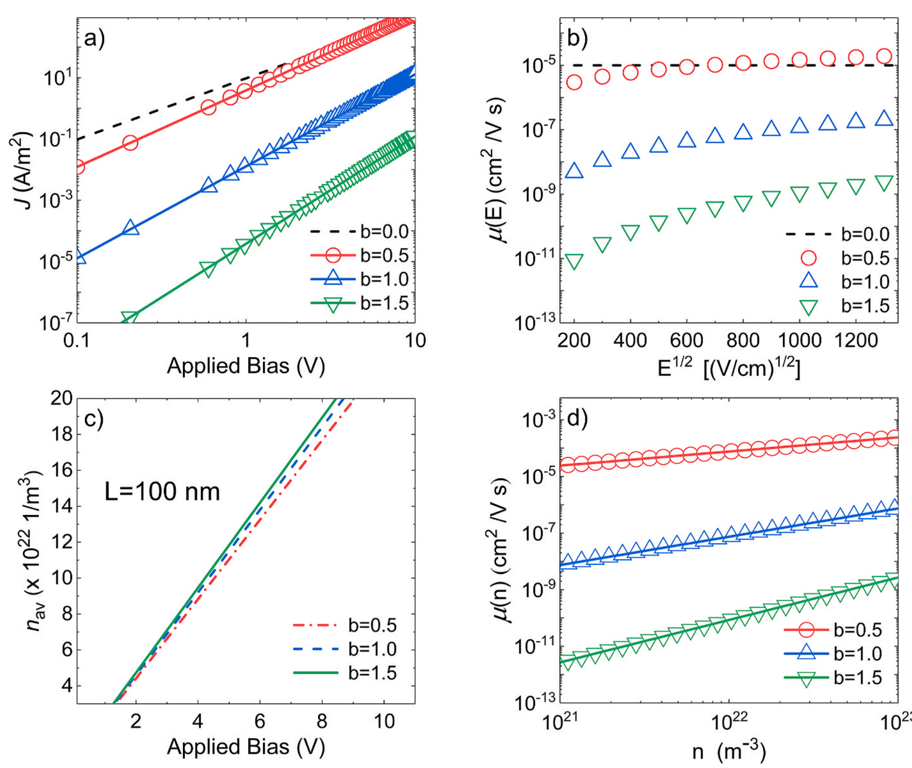

Fig. 5. (a) *J-V* curves for different b calculated with Equation (6). For different b, the corresponding a are calculated using Equation (18). (b) The calculated field-effect mobility using Eq. (18). (c) Average concentration n_{av} versus V derived from Equation (19), with L of 100 nm. (d) The EDOS mobility versus average concentration for various b calculated with Equation (18).

$$n_{av} = \frac{\int_0^L n(x)dx}{L} = \frac{\int_0^L \frac{\varepsilon}{q}\frac{(b+1)(2b+3)}{(b+2)^2}\frac{V}{L^2}\left(\frac{x}{L}\right)^{-1/(b+2)}dx}{L} = \frac{\varepsilon}{q}\frac{(2b+3)}{(b+2)}\frac{V}{L^2}. \tag{19}$$

For $b = 0$, $n_{av}=1.5(\varepsilon^*V/qL^2)$. This result is in agreement with the average density derived by C. Tanase et al. [22]. This demonstrates that for a single-carrier device the average current density depends quadratically on the voltage for low electric fields, as shown in Fig. 5c. Changes in the value of b at the same bias do not significantly affect the value of the average density.

The relationship between mobility and carrier concentration can also be derived directly from Eq. (18) as illustrated in Fig. 5d. This suggests that the carrier mobility is mainly determined by the carrier concentration. Similarly, the mobility decreases by several orders of magnitude with increasing b value while the density dependence is elevated. And it might indicate that the field dependence of the mobility, i.e., the so-called Poole-Frenkel effect, originates from the charge-density dependence of mobility.

To illustrate how to extract mobility from experimental data using our model, typical experimental data of devices from both small molecular (Ref. [9]) and polymeric (Ref. [11]) materials are selected. N. D. Nguyen et al. measured the *I–V* curve of ITO/α-NPD/Al device (see Ref. [9]) and extracted mobility values under different voltages from the peak frequencies of -Δ*B* curves. Due to the difference between the work functions of ITO (4.9 eV) and Al (4.3 eV), a built-in voltage $V_{bi}\approx0.6$ V was used to correct the *I–V* curve. Fig. 6a shows the corrected *I–V* data (see Ref. [9]) with a fitted slope of 2.5. This constant slope range (3–10 V) indicating that the mobility should be dominated by the carrier density. Therefore, we interprate the mobility values using Eq. (17) with peak frequencies of -Δ*B* curves (see Ref. [9]) and device thickness of 1340 nm. Fig. 6b gives the carrier mobility as a function of carrier density with a fitted slope of $b = 0.33$. The difference of the b values obtained from slope of *I–V* curve ($b = 2.5\text{-}2 = 0.5$) and slope of μ-n curve may stem from the injection barrier of hole from ITO into α-NPD (HOMO of 5.5 eV).

Fig. 6c exhibits the *J-V* curves of hole only devices made of glassy-poly(9,9-dioctylfluorene) (PFO) and β-phase PFO measured by X. Shi et al. (see Ref. [11]). As suggested by the authors, a $V_{bi} = 0.47$ V was used to correct the *J-V* curves. It can be seen that the *J-V* slopes are greater than 2 for both materials. X. Shi et al. measured the mobility of each of the two materials using the -Im*Z* method. Similarly, constant slope range (0.5–3.5 V) indicating that the mobility should be dominated by the carrier density. Fig. 6d illustrate the calculated μ using Eq. (17) versus carrier density, where the peak frequencies for calculating μ are extracted from Ref. [11]. Solid and dashed lines Fig. 6d indicate the fitted $\mu\sim n^b$ relations with $b = 0.44$ and 0.66 for glassy and β-phase PFO respectively. These b values are consistent with that fitted from *J-V* curves, suggesting that our model is competent for extracting mobility values for both small molecular and polymeric materials.

In order to use our EDOS model to interpret experimental results, several aspects should be considered. First, parameter b obtained from the slope of current density will be easily confused with the power of trap limited current density. The exist of trap distribution will cause an overestimation of parameter b [37] and a shift of the maximum frequency. Second, the density dependence of mobility, i.e., $\mu = an^b$, only

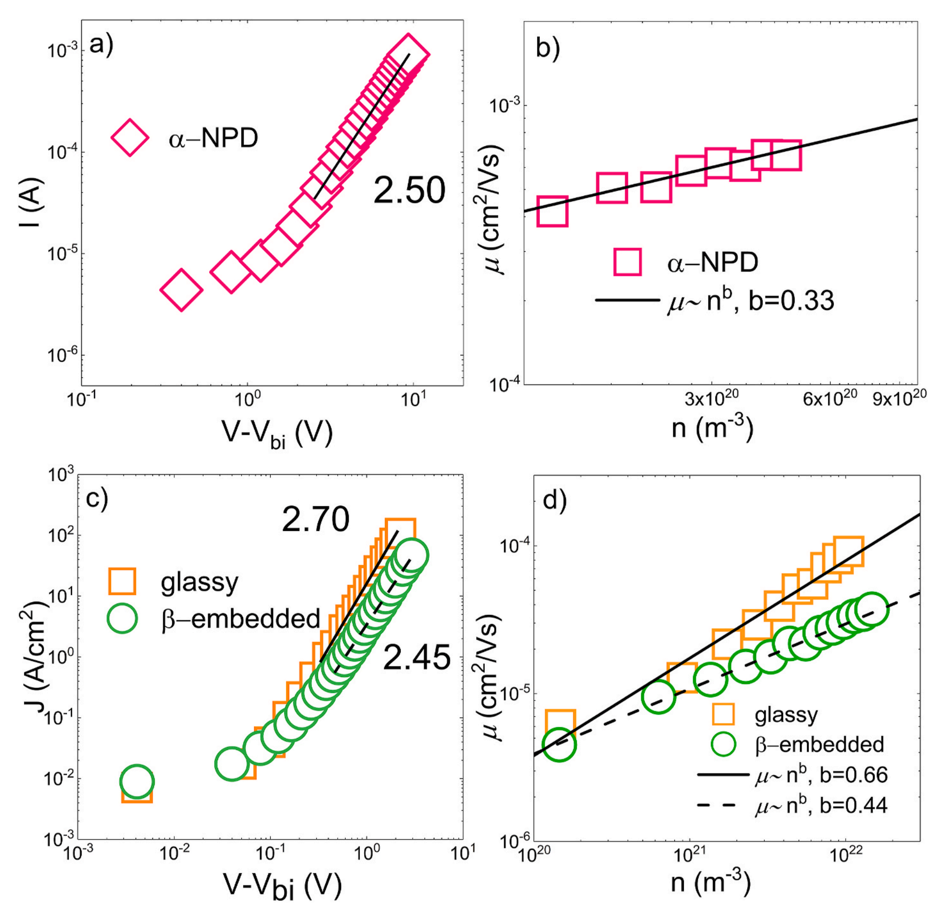

Fig. 6. (a) The *J-V* plot of ITO/α-NPD/Al device, extracted from Ref. [9]. This experimental data was corrected with a built-in voltage $V_{bi} = 0.6$ V. Solid line indicates the fitted slope. (b) The squares exhibit the mobility of α-NPD calculated by Eq. (17) using f_{max} obtained from Ref. [9]. Carrier density n is calculated by Eq. (19). The solid line is calculated $\mu\sim n^b$ with $b = 0.33$. (c) The *J-V* plots of glassy-PFO and β-phase PFO measured by X. Shi et al. in Ref. [11], which has been corrected with $V_{bi} = 0.47$ V. Solid line and dashed line indicate the fitted slopes. (d) The mobility of glassy-PFO and β-phase PFO calculated by Eq. (17) using f_{max} obtained from Ref. [11], Carrier density n is calculated by Eq. (19). The solid line and dashed line are calculations of $\mu\sim n^b$ with $b = 0.66$ and 0.44, respectively.

valid for EDOS conditions. For the case of Gaussian DOS, the mobility will be constant at low carrier density. Third, dispersive parameters should be included when dispersive transport involves.

4. Conclusion

In conclusion, the carrier density-dependent mobility of the EDOS model is investigated based on the work of B. Ramachandhran et al. The mobility calculated with the EDOS model compared to the classical MG model has a non-negligible error due to the presence of the exponential factor *b*. Neglecting the dominant contribution of carrier density to mobility may lead to the overestimation of field-dependent effect on mobility.

CRediT authorship contribution statement

Zhendong Ge: Writing – original draft. **Lei Wang:** Supervision. **Dawei Gu:** Validation. **Tianyou Zhang:** Conceptualization, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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